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Project Report

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Systems

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**Dementia Classification with Deep Neural Network using PET and MRI
images.**

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Abstract—This paper focuses on the use of convolutional neural network (CNN) to build and train a model for classification of dementia from non-dementia patients. The data involves the combination of magnetic resonance imagery (MRI) data and 3 different bio-tracer based positron emission tomography (PET) image data from the Open Access Series of Image Studies (OASIS) 3 project. The model was designed using layered structure of convolution, pooling and fully connected neurons. The training process includes cross-validation with a k value of 10. The generated model was tested on each data type, a high F1 score of 82% was achieved on the AV45 being the most reliable data type of the 4 used while the MRI was the least. The results further show the superiority of PET image data in classification of dementia from non-dementia patient opposed to the use of MRI based studies which has been the focus most projects reviewed. The project concludes with a recommendation on the effect of 1x1 kernel which should be further studied. For future work, focus must be made on the level of noise in the input images.

Keywords—CNN, F1-score, PET, MRI, OASIS3.

I. Introduction

Machine learning algorithms provides computers with the ability to find complex patterns in both structured and unstructured data such as images and recorded texts without specifically hardcoding them [4]. This method of generating insight has found profound application in various fields and has improved our quality of live. It has also proven beyond doubt its ability to aid medical experts in the diagnosis and prognosis of several diseases and possibly the stage of a disease in a medical subject to a very comparable accuracy thereby reducing human and systemic errors through the process [5][1][2]. Many authors have published several papers on the subject using several methods and different forms of data to achieve the same objective. This data includes the use of Electrocardiogram (ECG) however, most have used the magnetic resonance images MRI for analysis [6]. The traditional process of diagnosis includes manual inspection of CT, MRI and/or PET images by experts [3]. Earlier publications have resulted in the use of different methods for analysing and optimising how these diagnoses are carried out for better performance and assured safety and hence achieved different results. Many however also suffer from some limitations which will be reviewed in the next section for the purpose of advancement.

II. Literature Review

There has been an increase in the number of recorded dementia cases, and this is attributed to the fact that people now live longer than previous years [21] thereby increasing research data. It also presents researchers with the opportunity to research

methods of assessment and diagnosis [13]. These previous methods according to Ahmed et al (2019), who reviewed 90 previous related publications which spans the early era of diagnosis to the most recent, features methods of visual examination for human misjudgements to computer aided decision making. This increase in the number of diagnoses has put more burden on the experts by the volume of cases being reported against the available number of experts available to treat each case. Advancement in technology has helped improved these methods as neuroimaging technologies such as Magnetic Resonance Images (MRI), Positron Emission Tomography (PET) and the Single-Photon Emission Computed Tomography (SPECT) have been introduced. The next paragraph briefly introduces these technologies.

The neuroimaging technologies helps experts in the early detection of the pathology, its treatment and management which are important for preservation of life. Brain imagery is performed with different scanners depending on the purpose for examination which could be for either of the following assessments:

- Internal Structure
- Internal Functions
- Hybrid (Structural/Functional)

Oftentimes medical experts examine the brain structure for some abnormality and to ascertain the presence or absence of a certain disease. This is oftentimes done using the CT, or MRI however, when the need to examine the brain functionality and activeness, the PET or SPECT scans are done. There are also possibilities of combining the two modes for better insights. Some of these images have been made available for research purpose through some archives such as the ADNI and the Open Access Series of Imaging Studies (OASIS).

The brain MRI images have been profoundly used for dementia diagnosis largely due to the ease of differentiating between shades of grey and white matter in the image, they are generated in high resolution and less noise [11]. This has permitted the use of statistical methods to quantify the volume of grey matter against white matter in the brain thereby providing a reason for judging the presence dementia. Ahmed et al (2019) revealed that machine leaning approach: deep leaning, outperformed other methods in classification of dementia. This is exemplified in Marwa et al (2022) who used a CNN architecture and MRI dataset for classification with a testing accuracy of 99.68%. This result shows the high performance attainable with neural networks. In Marwa et al (2023), deep learning framework

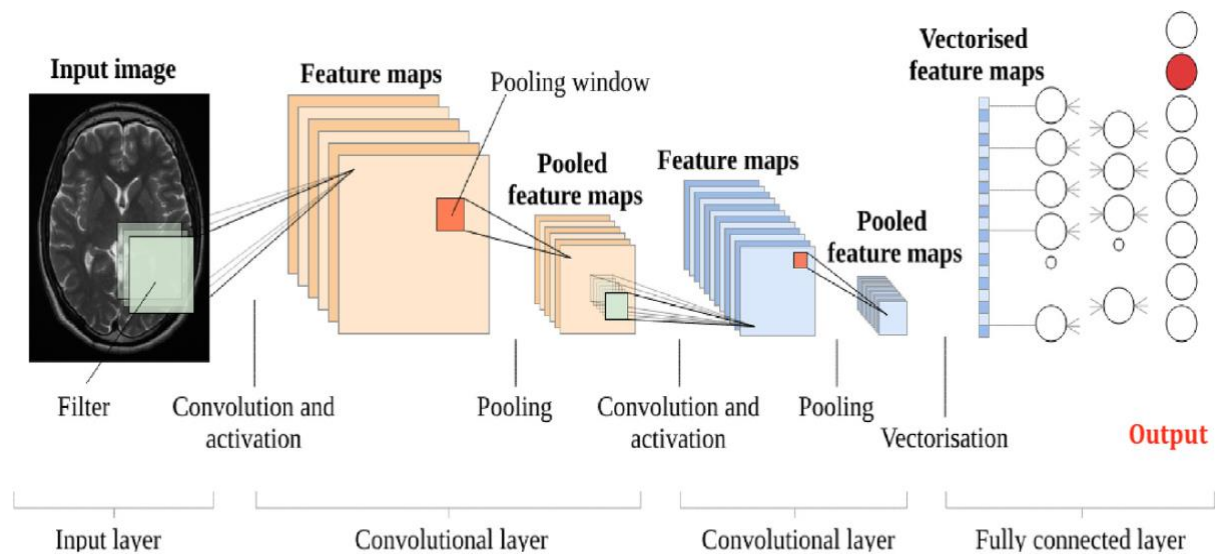


Figure 1 Convolution Neural Network Architecture

with boosting was also used, this helped with the improvement of the classifier and hence an overall accuracy of about $74\% \pm 9.487$, the data was a combination of MRI, PET and CSF, low amyloid beta 1-42 imaging in cerebrospinal fluid which contributed more to the analysis.

The PET imaging uses bio-tracers injected into the subject's blood, the blood flow is then monitored, allowing other body substances such as glucose and oxygen visible. This allows the consumption of oxygen and glucose to be monitored which in turn exposes brain activities in such region and hence reveals the functioning of the section of the brain [12]. The usefulness of the PET scans was established in Mathieu et al (2018) where PET was used to distinguish between different types of dementia [7]. Most related studies have only considered data from one bio tracer or another. Iaccarino et al (2017) and Subramanyam et al (2023) have shown that the combination of samples from different tracer-based scans can improve classification between different types of dementia and help improve the detection. Lorking et al (2021) concluded in his publication that the combined use of PET/MRI can improve insight into of the anatomical and functional changes in the brain which is needed for dementia detection and classification and ultimately for clinical management of these subjects. This in turn reveals that limited data from MRI can be augmented by PET, a practice which is manually practiced in the industry. The next section dives deep into the

strengths and weaknesses of previous papers and elaborates on what needs to be done.

Analysis

The field of data science generally suffers from insufficient data thereby reducing the trust associated with the analysis or model [14]. The papers reviewed in the previous section are no exemption. This analysis considers the strength and weaknesses of recent papers with a focus to build on the loopholes.

One of the most recent reviews on the use of PET/MRI for clinical diagnosis of dementia is Lorking et al (2021) who reviewed several papers between 2016 to 2021. These studies focused on the use of mixed-mode methods (PET/MRI) to achieve various aims which include classifying different types of dementia. Most of these studies were limited in the number of patients which were included in the study. The methods used in these papers includes classification algorithms as Random Forest (RF), Support Vector Machine (SVM) and k-Nearest Neighbour as opposed to the use of deep learning. The review shows the importance of PET/MRI images in detection but suffers from insufficient data [12]. To compensate for insufficient data this paper presents the use of different bio tracer-based PET images with MRI for improved performance.

While many researchers have published several papers on the topic using machine learning framework, the methods used by these researchers

are oftentimes not repeatable due to the use of third-party software for image pre-processing thereby creating an inconsistent data input for the machine learning model [15]. This also makes it unimplementable for the end users, medical experts, as they must integrate several software to achieve the same result.

During the planning stage, several decisions such as technology which will be used for processing the images, and how the image pre-processing would be done had to be dropped as it was not in the favour of reproducibility. While most of the reviewed papers have used one custom processor or the other, this paper seeks to use a commercially available processor from Amazon Web Services through Sagemaker which is able to process infinite volume of data thereby increasing ability to train more data.

This study is based on the use of CNN algorithmic method for classification. This leads to the description of the method, which is comparable to ANN, a method made up of neurons with updatable weights and biases such that a training data is used to adjust the weights and bias tensors iteratively through a classifier function. A correction system using a differentiable score and loss function is used to track the progress of the process. The whole process of adjusting the weights and bias tensor using a pre-processed input is called training. The process passing the $N \times N$ input data through a set of connected layers which could be either of the following:

- Convolution Layer
- Pooling Layer
- Fully Connected Layer

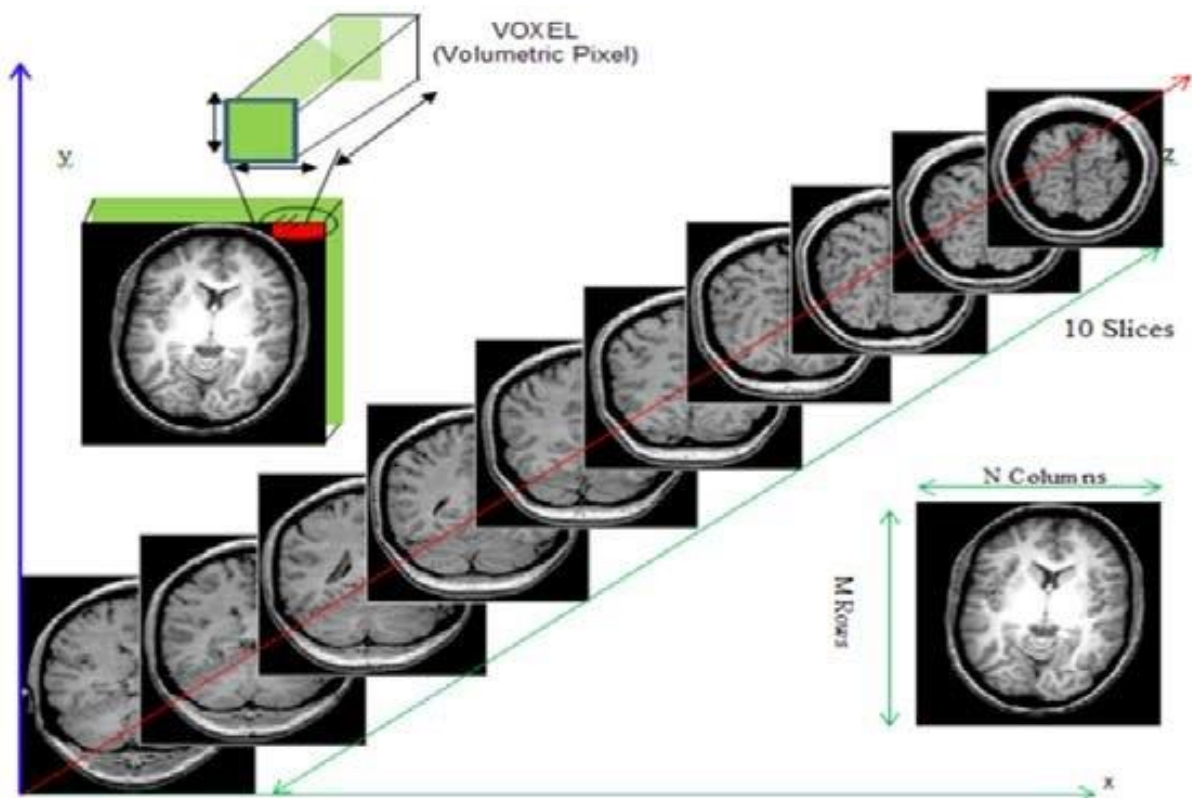


Figure 2 3D slices of MRI images

This also presents us with models from other researchers for comparison.

III. Methods

A. Formulation

Each layer is responsible for one form of processing on the input layer, its output is then forwarded to the next layer for further processing. This forms a nonlinear network architecture which is unique to each model. This nonlinearity is achieved through an activation function which is extensively discussed in later sections.

This can be expressed mathematically as follows:

$$Y = A(W^1X^1 + W^2X^2 + W^nX^n \dots + B^n) \quad (1)$$

n: size of input, X: input features, W: weights, A: activation function

B. Data Selection and Description

The OASIS dataset prepared by the Washington University Knight Alzheimer Disease Research Centre in several ongoing studies [220]. The data consists of 1098 participants of which 605 are cognitively normal adults, and 493 at various stages of cognitive decline with an age range of 42 – 95 years. The data contains about 2000 MRI sessions and over 1500 raw PET imaging sessions. The data also contains details regarding their respective dementia rating, age, etc.

image and natural processing studies). Oftentimes, the unavailable data or high cost related to sourcing for more data makes the data augmentation a key method in most related studies. They help in reducing overfitting.

Following from the various studies reviewed in previous chapter, it was established that most of them suffered from unavailable and limited data thus impacting the performance of the models used. This created a need for increase in available data necessary for training and testing. The data was downloaded in a 3D and 4D NIFTI file format from the archive and required transformation to a NumPy array from which images can be extracted, standardized, and cleaned. The process of cleaning, standardizing, and cleaning accounts for about 50% of the time spent on the project as some required visual inspection for correct decisions to be made. To generate enough images from the large volume of brain imagery available it is crucial to take to advantage of the numerous numbers of slices from the 3D or 4D array. It was determined that since there are a minimum of 36 image slices cutting

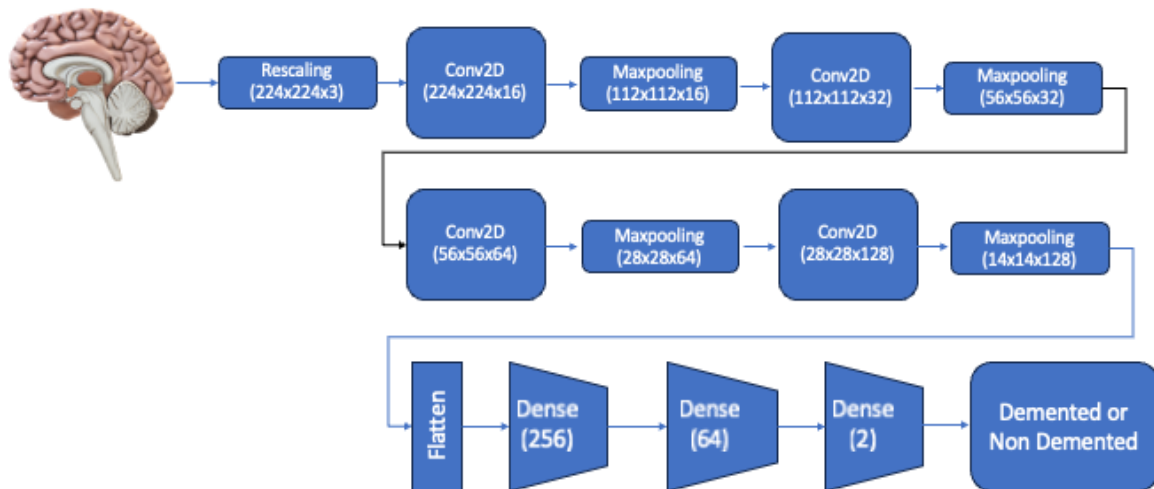


Figure 3 Figure 4 Proposed CNN layer design.

C. Data Augmentation and Cleaning

Data augmentation is the process of artificially increasing the number of available data by generating new data from the existing data which could be achieved by carrying out minor alterations to the original data or creating more variation of the same data using machine learning (ML) within the sample space. They find a great importance in studies where available data is limited (typical to

through the length of the brain being studied, more images slice closest to the hippocampus has the best probability for the study and detection of dementia [20]. Hence an average of about 10 images was subtracted from each imaging files array as opposed to most study which considered only the use of one image slice from the subject. This provides a comparable more ground truth data per session for training and testing purpose. The MRI files after conversion was a 3D (L, W, H) array as shown in fig 2 below, W and H representing the width and height

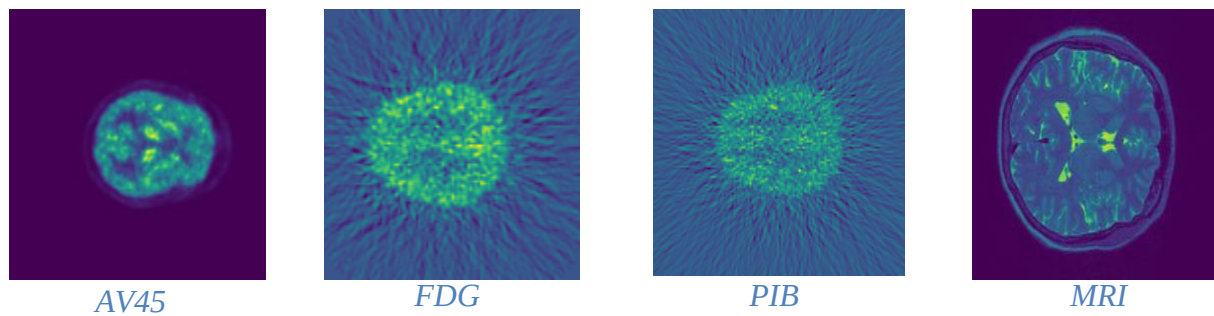


Figure 4 Sample of each image type

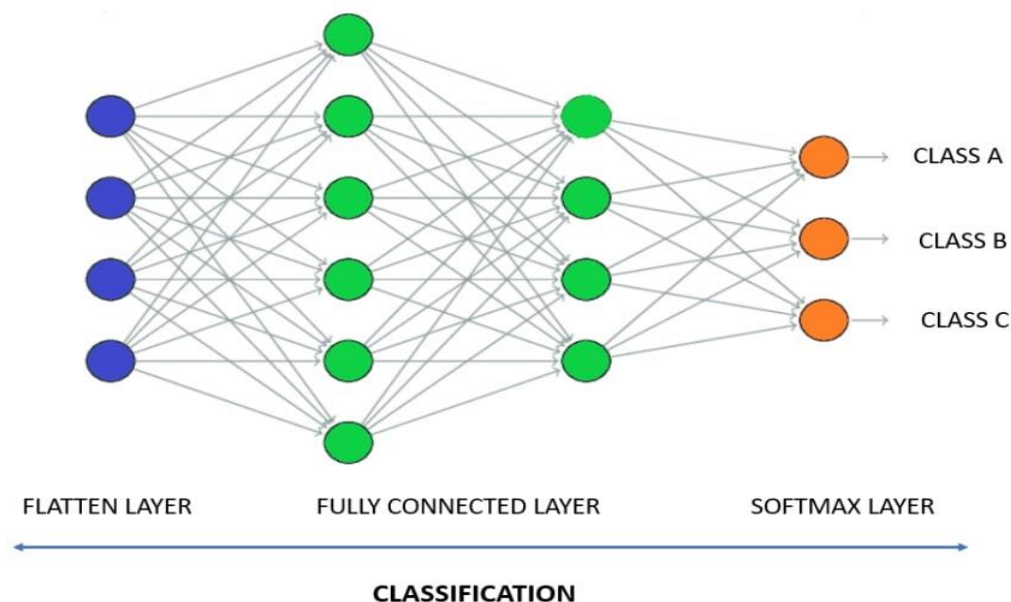


Figure 5 CNN Fully Connected Layers

of the image respectively and L representing the number of slices along the length of the brain while the PET scans were converted to a 4D (T, L, W, H) array, T representing time frame.

The literature review reveals that there are more studies that have used MRI images which has generally not been able to achieve a reliable result in classifying demented from nondemented patients. The limited availability of ground truth image data has led to the use of PET images which has found success in many recent studies. Hence, this study uses a combination of PET and MRI images. We also took a further step in combining different PET image types namely, AV45, PIB and FDG which are

short names for bio-tracers used by the scanner. PET images generally contain a higher level of noise compared to MRIs as their principle of operation considers contrast in illumination in the provided image to identify functional section of the brain and this contribute to their limitation as opposed to MRIs which could differentiate between classes of dementia by the presence of a different shade of grey. It is therefore a necessity to carry out some form of noise reduction for a better result which falls

out of the scope of this project as there is a limited amount of time and resources available.

D. Network Architecture and Strategies in pre-processing

image, its size and shape determines the output. The network used contains 4 convolutional and maxpooling layer. After many trials a 1x1 kernel has been used on the convolution layer to reduce the channel depth while activating each with a rectified linear unit (ReLU) and the maxpooling layer with a

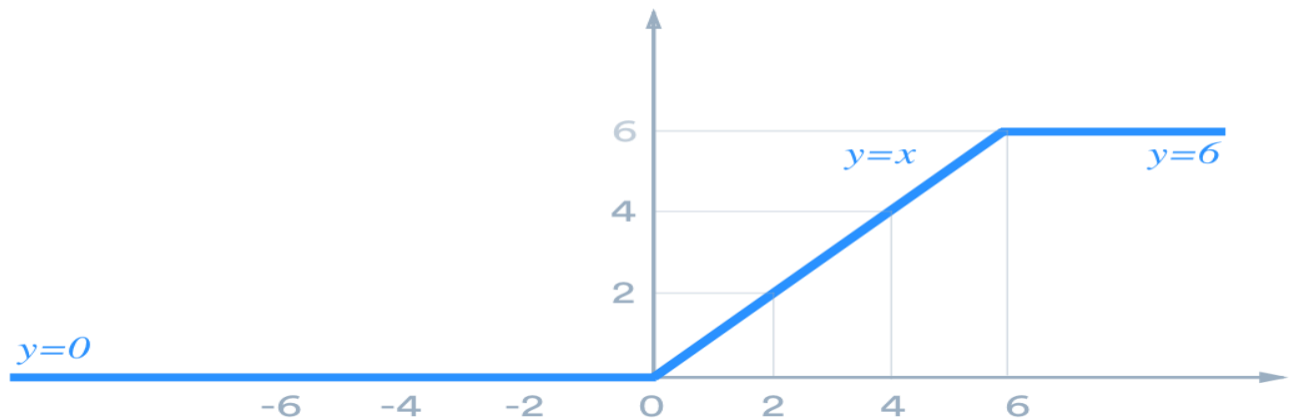


Figure 6 Rectified Linear Unit Function

Our proposed network above is a deep convolutional neural network which was carefully designed to achieve a high, reliable, and repeatable performance by varying various hyperparameters. This section discusses the design process and various importance of some key hyperparameter. The network consists of various connected layers as shown in figure 4 above, which could be either convolution, pooling and fully connected layer. The network starts with normalising the input data presented to the yet to be designed network, which involves ensuring the mean and standard deviation is zero and 1 respectively. This helps to reduce the skewness of the data while also making them uniform, thereby stabilizing the gradient descent and allowing for faster learning rate hence faster training process. To add variability to the data, which was overfitting at the first try, data augmentation layer was implemented on each batch data. The data augmentation layer combines for zoom, rotation, flip and resizing, hence reducing overfitting. Several parameters tuning which involves degree of zoom, angle of rotation etc. was carried out on the data augmentation layer to produce an acceptable level of variability.

The convolutional layer forms a very important layer as its stand to be the layer which extracts features from the data. This process is carried out using the convolution operator to convolute the image and stores the result in a channel. The kernel serves as the operator which is used to convolve the

shape of 2 was used to pool the data thereby achieving dimension reduction. The reason for the use of such small kernel size is justified by the number of features needed to be identified by the model to reliably classify the data.

$$\text{ReLU formula is } f(x) = \max(0, x)$$

The last set of layers include the fully connected layer which linearises the input 3D data and is fed directly into the connected neurons. These connected neurons run deep as feature a very deep interconnectedness between one another. An output is ultimately generated from each dense layer which is then passed on to the next hence, yielding a prediction and in this case binary.

The network is then used to learn a weight which we then use to predict on the test set. This then leads us to the next chapter which discusses the results achieved by the model.

IV. Results and Discussions

A. Experimental Setting

The proposed model was implemented using TensorFlow, Keras, and Python on a Linux machine hosted on the AWS EC2 P3 instance which features

up to 8 NVIDIA V100 Tensor Core GPU with 100Gb network throughput. This made the processing of such large data possible compared with CPU only based computers which crashed abruptly during training. A CNN training with a mini batch of 32 was applied with “Adam” optimizer (Default parameters were used). To achieve a validated result, Random Forest algorithm, which is self-learning and operates by ensembling various decision trees, was proposed, and implemented on the same computer and data. Training and testing were also carried out on each type of dataset to record model performance against joining the data. This helps measure trade-offs in terms of precision and recall.

B. Performance Metric

To start with, it is necessary to discuss the balancing of the data used and how the classes were chosen. To avoid data imbalance, it was necessary to have make equal the number of images from both classes. It is however needful to understand that each data type (i.e., AV45, FDG, MRI and PIB) have different number of images generated, the MRI accounts for most of the data and it is worth noting that the data presented in the OASIS3 project was already classified according to Clinical Dementia Rating (CDR) but ranges between 0.0 – 3.0. It was decided that the classes 0.0 and 0.5 be considered for the classification to simplify the amount of data that would be used and noting that there are more data for the two classes and would hence be advantageous to the study. This however increased the recall required from the model in that it is harder to spot a difference between the two classes. The lower limit for the number of available images was picked for the balancing of the classes for each data type thereby generating a balanced dataset for training.

For quantitative evaluation of the results four metrics have been used which includes accuracy, precision, recall/sensitivity, and F1-score. We start by defining the following terms which are fundamental to the derivation of the above metrics.

- i. True Positive: This is an outcome of correctly predicted positive class by the model.
- ii. True Negative: This is also the outcome of correctly predicted negative class by the model.
- iii. False Positive: This is an outcome for which the model incorrectly predicts the positive class.

- iv. False Negative: This is also an outcome in which the model incorrectly predicts the negative class.

Accuracy: This is the fraction of rightly predicted samples to the total number of predictions. This metric does not have the ability to inform on the performance of the model when there is imbalance in dataset. While precisions talk about the proportion of positively predicted which was correct.

- $$\text{Accuracy} = \frac{(TP+TN)}{(TP+TN+FP+FN)}$$
- $$\text{Precision} = \frac{TP}{TP + FP}$$
- $$\text{Recall} = \frac{TP}{TP + FN}$$
- $$\text{F1 - score} = \frac{2TP}{2TP + FP+FN}$$

Recall defines the proportion of actual positives which was correctly identified. It is worth understanding that precision and recall are connected such that an increase in one lead to a decrease in the other. Hence, considering the nature of this research, recall is the key performance metric to watch out for. Hence we must try to increase the number of True positives while reducing the False negatives.

C. Results

The classification result from the proposed CNN and RF model is reported above. As earlier stated, the Random Forest was used for comparison. The evaluation process begins by ensuring that overfitting did not occur through the training process. The early round of training features some underfitting which was overcome by reducing the kernel size to 1x1, a case to be studied. It is believed to have enhanced the model to take more focus on the minute details in the images to distinguish each class from another. The progress of the training process is monitored by comparing the accuracies and losses of training and validation data. A higher validation accuracy signifies the presence of overfitting which means the trained model performs better on new data against those trained on. It could also signify that the model complexity in learning on patterns is more than required. The data is split into K-fold, the Kth fold is used as a validation while the K-1 folds are used as training the process of splitting and validating the model is known as cross-validation. This model uses the default k = 10 from TensorFlow. Figure 7 is a plot of the training and validation accuracies and losses to confirm the absence of overfitting or underfitting, a similar occurrence to overfitting, in which the model is unable to learn and predict on training and validation data respectively. The overall training accuracy was 77.88% while the validation accuracy was 75.81%,

while the losses were also reduced from about 0.7 to 0.45.

The CNN model was tested on each data type after training, this helped measure how efficient the model performed on each data type. The tables below summarise the result of the trained model on each data type. Table 1 presents the classification result on testing with the PIB tracer PET data. Since this project is related to medical diagnosis, high recall is required to ensure reliability however since we compared the output over various data types we consider the F1 score which can combine precision and recall as hence provide a basis for comparing the tables. The PIB result features the lowest score of 66% and 39% for non-demented and demented respectively. The 66% for the nondemented class shows the ability of the model to identify this class while it is inefficient to pick the required patterns in a demented image making it the worse performed data type of the four. Table 2 presents the results for the AV45 tracer PET data. With an F1 score of 82% and 81% for the respective classes, this data shows the most reliability result of the four. This is evident in the visual representation of this image type which contrasts to the blur image in the PIB type. The FDG result is presented in Table 3 where it shows an F1

score of 75% and 71% for the classes respectively, a good result with an acceptable reliability which can be improved further. Table 4 represents the MRI classification results the 73% and 69% for the respective classes which is also acceptable. In addition, it is observed that the F1 scores from the PET image types except the PIB are generally better than the MRI and the model has a better recall on the non-demented class.

It is evident that the model ultimately achieved a good result while it failed on other data types which invariably suggests further research and processing on such data. The AV45 Pet images achieved highest F1 score suggesting a higher reliability compared to others. The FDG PET and MRI have also shown to produce a reliable result with a range of 69% – 75%. These image types (AV45, FDG and MRI) also features a considerable high recall which is required since misclassification could be very costly and not acceptable. Finally, it is worth noting that the RF model failed on the testing as a very low F1 score of 21% - 24% was achieved. This will be further discussed in the next section.

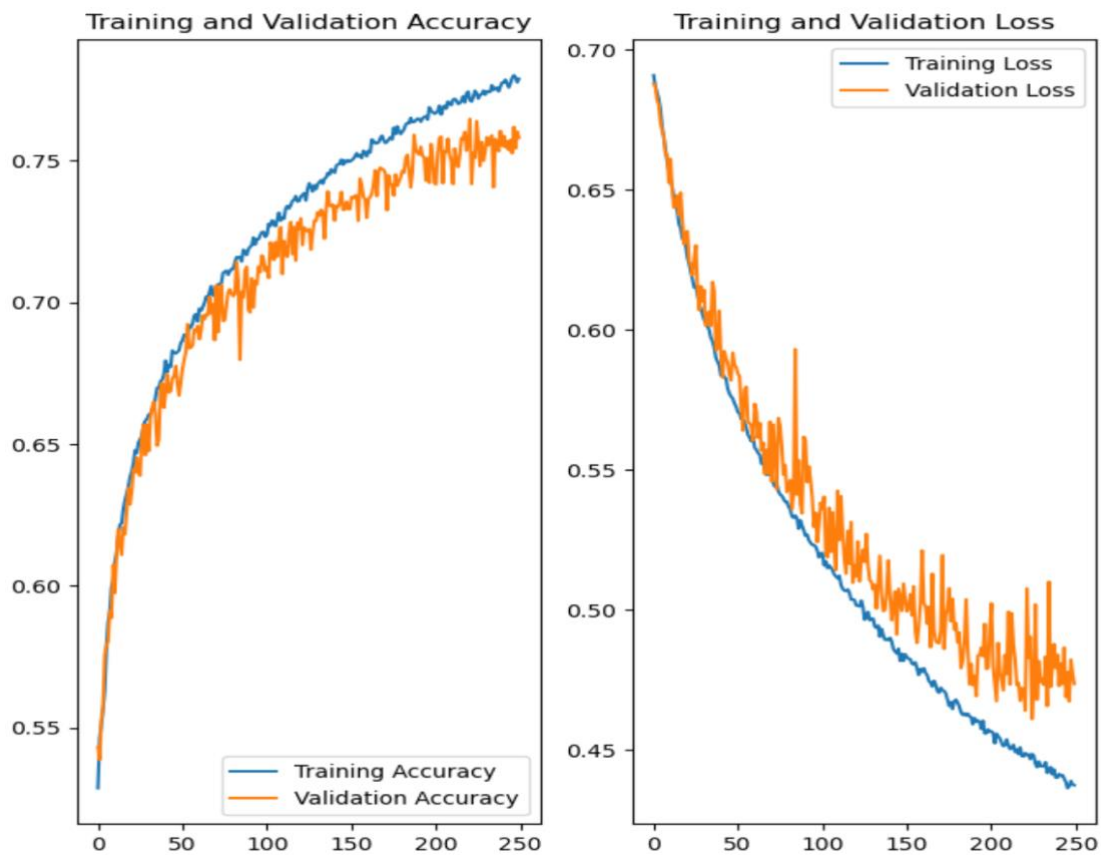


Figure 7 Training vs Validation Accuracy and Loss

Table 1 PIB result

No	Precision	Recall	F1-Score	Support
Non-Demented	0.54	0.85	0.66	549
Demented (0.5)	0.65	0.28	0.39	549

Table 2 AV45 Result

No	Precision	Recall	F1-Score	Support
Non-Demented	0.79	0.86	0.82	3150
Demented (0.5)	0.85	0.77	0.81	3150

Table 3 FDG result

No	Precision	Recall	F1-Score	Support
Non-Demented	0.71	0.80	0.75	248
Demented (0.5)	0.77	0.67	0.71	248

Table 4 MRI result

No	Precision	Recall	F1-Score	Support
Non-Demented	0.69	0.77	0.73	1277
Demented (0.5)	0.74	0.65	0.69	1276

Table 5 RF MRI result

No	Precision	Recall	F1-Score	Support
Non-Demented	0.24	0.24	0.24	1277
Demented (0.5)	0.21	0.21	0.21	1276

V. Conclusions and Recommendations

Other projects have used MRI in conjunction with only one PET tracer-based image data however, this project has successfully used a combination of multiple (3) biomarker-based PET image in conjunction with MRI to build, train and test a CNN

model which has helped with the following conclusions and recommendations.

The combination of MRI images with different biomarkers-based PET images has proven to improve training process and helped the overall model increase in its prediction hence these other

image type (PET) can be used to augment during training against the use of a single bio-maker PET image type in conjunction with MRI. It can also be concluded, since the F1 scores from the PET images except the PIB are higher than the MRI, that the PET image data are better in predicting dementia compared to MRI image data type.

The results in the previous section have evidently shown that images produced from FDG and AV45 scanners have more reliability compared to others. This suggests that the presence of illumination in these PET image types can be used by the model to distinguish between the classes. A sample from each image type is shown in figure 3 under the data augmentation section for visual comparison. This leads to the recommendation that PIB based images from scanners would require further processing aimed at reducing noise and increasing the illumination denoting brain activity before being used for training.

Finally, it is important to carry out research to evidently understand the reasons why 1x1 kernel size has given a better result compared to 3x3 kernel size earlier used during the CNN training. It is also recommended that other machine learning algorithms such as Generative Adversarial Networks GAN are used to produce augmented images.

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